

driver	unknowns	discrete model	solve and certificate
transient	$[n, p, P^+, (P^-)]$	SG carrier/ion continuity Poisson at each RHS call	Radau IIA history-preserving state
steady_state	$[\ln n, \ln p]$	1D density RHS fixed ion profile	direct Newton DC residual checks
quasi_fermi	$[\Delta\varphi_n, \Delta\varphi_p]$	eliminated Poisson response optional local interface solve	Newton continuation cell, current, Poisson certificates
QF frequency	complex small signal	linearization about a certified QF operating point	refined complex solve all-face admittance continuity
2D	$[n(y, x), p(y, x)]$	2D SG + sparse Poisson static ion background	Radau-LU lateral/vertical diagnostics

Driver choice fixes the unknowns, supported physics,
and required certification checks.